NAME: ALEJANDRO PANTOJA-ZURITA

CLASS: CSC 210 – SYSTEM PROGRAMMING

ASSIGNMENT: MODULE 2 – LAB 2 – LINUX COMMAND PRACTICE

INSTRUCTOR: TIAN Z. DING.

DATE: SEPTEMBER 3, 2025

LINUX COMMANDS

1) “ls” command - Lists the contents of a directory (files and subdirectories)

```
alejandr0@raspberrypi:~ $ cd Downloads
alejandr0@raspberrypi:~/Downloads $ ls
test.txt
alejandr0@raspberrypi:~/Downloads $ cd
alejandr0@raspberrypi:~ $ ls
Bookshelf  Documents  Music     Public     Videos
Desktop    Downloads  Pictures  Templates
alejandr0@raspberrypi:~ $
```

2) “cd” command - Changes the current working directory.

```
alejandr0@raspberrypi:~ $ ls
Bookshelf  Documents  Music     Public     Videos
Desktop    Downloads  Pictures  Templates
alejandr0@raspberrypi:~ $ cd Pictures
alejandr0@raspberrypi:~/Pictures $ ls -l
total 0
alejandr0@raspberrypi:~/Pictures $ cd
alejandr0@raspberrypi:~ $ cd Downloads
alejandr0@raspberrypi:~/Downloads $ cd Pictures
bash: cd: Pictures: No such file or directory
alejandr0@raspberrypi:~/Downloads $ cd
alejandr0@raspberrypi:~ $
```

3) “mkdir” (make directory) command - Creates a new directory.

```
alejandr0@raspberrypi:~/Downloads $ cd Pictures
bash: cd: Pictures: No such file or directory
alejandr0@raspberrypi:~/Downloads $ cd
alejandr0@raspberrypi:~ $ cd
alejandr0@raspberrypi:~ $ cd Downloads
alejandr0@raspberrypi:~/Downloads $ mkdir test_Folder_01
alejandr0@raspberrypi:~/Downloads $ ls
test_Folder_01  test.txt
alejandr0@raspberrypi:~/Downloads $
```

4) “rmdir” (remove directory) command - Removes an empty directory.

```
alejandr0@raspberrypi:~/Downloads $ mkdir test_Folder_01
alejandr0@raspberrypi:~/Downloads $ ls
test_Folder_01  test.txt
alejandr0@raspberrypi:~/Downloads $ rmdir test_Folder_01
alejandr0@raspberrypi:~/Downloads $ ls
test.txt
alejandr0@raspberrypi:~/Downloads $
```

5) “rm” (remove) command - Deletes files or directories. Use with caution.

```
alejandr0@raspberrypi:~/Downloads $ ls
test_02.txt  test.txt
alejandr0@raspberrypi:~/Downloads $ rm test.txt
alejandr0@raspberrypi:~/Downloads $ ls
test_02.txt
alejandr0@raspberrypi:~/Downloads $
```

6) “chmod” (change mode) command - Changes the permissions of a file or directory.

```
alejandr0@raspberrypi:~ $ touch script.sh
alejandr0@raspberrypi:~ $ ls -l
total 36
drwxr-xr-x 2 alejandr0 alejandr0 4096 Apr  4  2022 Bookshelf
drwxr-xr-x 2 alejandr0 alejandr0 4096 Aug 26 19:53 Desktop
drwxr-xr-x 2 alejandr0 alejandr0 4096 Aug 26 19:53 Documents
drwxr-xr-x 2 alejandr0 alejandr0 4096 Sep  2 20:26 Downloads
drwxr-xr-x 2 alejandr0 alejandr0 4096 Aug 26 19:53 Music
drwxr-xr-x 2 alejandr0 alejandr0 4096 Aug 26 19:53 Pictures
drwxr-xr-x 2 alejandr0 alejandr0 4096 Aug 26 19:53 Public
-rw-r--r-- 1 alejandr0 alejandr0    0 Sep  2 21:09 script.sh
drwxr-xr-x 2 alejandr0 alejandr0 4096 Aug 26 19:53 Templates
drwxr-xr-x 2 alejandr0 alejandr0 4096 Aug 26 19:53 Videos
alejandr0@raspberrypi:~ $ chmod +x script.sh
alejandr0@raspberrypi:~ $ ls -l script.sh
-rwxr-xr-x 1 alejandr0 alejandr0 0 Sep  2 21:09 script.sh
alejandr0@raspberrypi:~ $
```

7) “mount” command - Mounts a filesystem to a directory.

```
alejandr0@raspberrypi: ~
File  Edit  Tabs  Help
alejandr0@raspberrypi:~ $ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
sda           8:0    1   3.8G  0 disk /media/alejandr0/GORO
mmcblk0     179:0    0 119.4G  0 disk
├─mmcblk0p1 179:1    0   2.9G  0 part
├─mmcblk0p2 179:2    0     1K  0 part
├─mmcblk0p5 179:5    0    32M  0 part
├─mmcblk0p6 179:6    0   256M  0 part /boot
└─mmcblk0p7 179:7    0 116.2G  0 part /
alejandr0@raspberrypi:~ $ sudo mount /dev/sda /mnt/usb
alejandr0@raspberrypi:~ $ ls /mnt/usb
'System Volume Information'   test
alejandr0@raspberrypi:~ $ sudo umount /mnt/usb
alejandr0@raspberrypi:~ $
```

8) “cp” command - Copies files or directories.

```
alejandr0@raspberrypi:~/Downloads $ ls
test_02.txt
alejandr0@raspberrypi:~/Downloads $ cp test_02.txt test_01.txt
alejandr0@raspberrypi:~/Downloads $ ls
test_01.txt  test_02.txt
alejandr0@raspberrypi:~/Downloads $
```

9) “mv” command - Moves or renames files and directories.

```
alejandr0@raspberrypi:~/Downloads $ ls
test_02.txt
alejandr0@raspberrypi:~/Downloads $ cp test_02.txt test_01.txt
alejandr0@raspberrypi:~/Downloads $ ls
test_01.txt  test_02.txt
alejandr0@raspberrypi:~/Downloads $ mv test_01.txt test_01_Renamed.txt
alejandr0@raspberrypi:~/Downloads $ ls
test_01_Renamed.txt  test_02.txt
alejandr0@raspberrypi:~/Downloads $
```

10) “find” command - Searches for files and directories in a directory hierarchy.

```
alejandr0@raspberrypi:~ $ find . -name "*.txt"
./.pki/nssdb/pkcs11.txt
./.config/chromium/Default/Extensions/cjpalhdlnbpafiamejdnhcphjbkeiagm/1.65.0_0/assets/ublock/badware.min.txt
./.config/chromium/Default/Extensions/cjpalhdlnbpafiamejdnhcphjbkeiagm/1.65.0_0/assets/ublock/badlists.txt
./.config/chromium/Default/Extensions/cjpalhdlnbpafiamejdnhcphjbkeiagm/1.65.0_0/assets/ublock/unbreak.min.txt
./.config/chromium/Default/Extensions/cjpalhdlnbpafiamejdnhcphjbkeiagm/1.65.0_0/assets/ublock/filters.min.txt
./.config/chromium/Default/Extensions/cjpalhdlnbpafiamejdnhcphjbkeiagm/1.65.0_0/assets/ublock/quick-fixes.min.txt
./.config/chromium/Default/Extensions/cjpalhdlnbpafiamejdnhcphjbkeiagm/1.65.0_0/assets/ublock/privacy.min.txt
./.config/chromium/Default/Extensions/cjpalhdlnbpafiamejdnhcphjbkeiagm/1.65.0_0/assets/thirdparties/easylist/easylist.txt
./.config/chromium/Default/Extensions/cjpalhdlnbpafiamejdnhcphjbkeiagm/1.65.0_0/assets/thirdparties/easylist/easyprivacy.txt
./.config/chromium/Default/Extensions/cjpalhdlnbpafiamejdnhcphjbkeiagm/1.65.0_0/assets/thirdparties/urlhaus-filter/urlhaus-filter-online.txt
./.config/chromium/Default/Extensions/cjpalhdlnbpafiamejdnhcphjbkeiagm/1.65.0_0/css/fonts/Inter/LICENSE.txt
./.config/chromium/Default/Extensions/cjpalhdlnbpafiamejdnhcphjbkeiagm/1.65.0_0/LICENSE.txt
./.config/chromium/Default/Extensions/cjpalhdlnbpafiamejdnhcphjbkeiagm/1.65.0_0/web_accessible_resources/README.txt
./.config/chromium/Default/Extensions/cjpalhdlnbpafiamejdnhcphjbkeiagm/1.65.0_0/web_accessible_resources/noop.txt
./.config/chromium/Default/Extensions/cjpalhdlnbpafiamejdnhcphjbkeiagm/1.65.0_0/img/fontawesome/LICENSE.txt
./.local/share/Trash/files/test_02.txt
./.local/share/Trash/files/test_01_Renamed.txt
alejandr0@raspberrypi:~ $ find /etc -type d -name "ssh"
/etc/ssh
/etc/sv/ssh
find: ‘/etc/polkit-1/localauthority’: Permission denied
find: ‘/etc/ssl/private’: Permission denied
find: ‘/etc/cups/ssl’: Permission denied
alejandr0@raspberrypi:~ $
```

11) “paste” command - Merges lines of files side by side.

```
alejandr0@raspberrypi:~ $ echo -e "A\nB\nC" >fileA.txt
alejandr0@raspberrypi:~ $ echo -e "1\n2\n3" >fileB.txt
alejandr0@raspberrypi:~ $ cat fileA.txt
A
B
C
alejandr0@raspberrypi:~ $ cat fileB.txt
1
2
3
alejandr0@raspberrypi:~ $ paste fileA.txt fileB.txt
A       1
B       2
C       3
alejandr0@raspberrypi:~ $
```

12) “cut” command - Extracts specific columns or fields from a file or input.

```
alejandr0@raspberrypi:~ $ echo "apple, banana, cherry" > fruits.csv
alejandr0@raspberrypi:~ $ cut -d',' -f2 fruits.csv
 banana
alejandr0@raspberrypi:~ $ cat fruits.csv
apple, banana, cherry
alejandr0@raspberrypi:~ $
```

13) “sort” command - Sorts lines in a text file.

```
alejandr0@raspberrypi:~ $ echo -e "grape\napple\nbanana" > fruits.txt
alejandr0@raspberrypi:~ $ cat fruits.txt
grape
apple
banana
alejandr0@raspberrypi:~ $ sort fruits.txt
apple
banana
grape
alejandr0@raspberrypi:~ $
```

14) “join” command - Joins lines of two files on a common field.

```
alejandr0@raspberrypi:~ $ echo -e "1 Alice\n2 Bob" > file1.txt
alejandr0@raspberrypi:~ $ echo -e "1 USA\n2 UK" > file2.txt
alejandr0@raspberrypi:~ $ join file1.txt file2.txt
1 Alice USA
2 Bob UK
alejandr0@raspberrypi:~ $
```

15) “awk” command - A powerful text processing tool used for pattern scanning and processing.

```
alejandr0@raspberrypi:~ $ echo -e "John 25\nJane 30" > people.txt
alejandr0@raspberrypi:~ $ awk '{print $1}' people.txt
John
Jane
alejandr0@raspberrypi:~ $ awk '$2 > 26' people.txt
Jane 30
alejandr0@raspberrypi:~ $
```

16) “touch” command - Creates an empty file or updates the timestamp of a file.

```
alejandr0@raspberrypi:~ $ touch newfile.txt
alejandr0@raspberrypi:~ $ ls -l newfile.txt
-rw-r--r-- 1 alejandr0 alejandr0 0 Sep  2 21:01 newfile.txt
alejandr0@raspberrypi:~ $
```

17) “|” (Pipe) operator – Passes output of one command as input to another.

```
alejandr0@raspberrypi:~ $ ls | grep ".txt"
file1.txt
file2.txt
fileA.txt
fileB.txt
fruits.txt
newfile.txt
people.txt
alejandr0@raspberrypi:~ $
```

18) “<” (input redirection) operator – Takes input from a file.

```
alejandr0@raspberrypi:~ $ sort < fruits.txt
apple
banana
grape
alejandr0@raspberrypi:~ $
```

19) “>” (output redirection) operator – Writes output to a file (overwrites).

```
alejandr0@raspberrypi:~ $ echo "Overwrite this" > output.txt
alejandr0@raspberrypi:~ $ cat output.txt
alejandr0@raspberrypi:~ $ echo "Overwrite this a second time!" > output.txt
alejandr0@raspberrypi:~ $ cat output.txt
Overwrite this a second time!
alejandr0@raspberrypi:~ $
```

20) “>>” (append redirection) operator – Appends output to the end of a file.

```
alejandr0@raspberrypi:~ $ cat output.txt
Overwrite this a second time!
alejandr0@raspberrypi:~ $ echo "Append this" >> output.txt
alejandr0@raspberrypi:~ $ cat output.txt
Overwrite this a second time!
Append this
alejandr0@raspberrypi:~ $
```